Digital Catalysis: Accelerated Discovery through Human in the Loop

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To meet the demand for new catalyst materials traditional one-factor-at-a-time (OFAT) methods fall short and systematic methods are needed.[1] Herein we propose a human-in-the-loop methodology that systematically integrates expert knowledge, model-based techniques, and Design-of-Experiments (DoE). This method maintains the adaptability and robustness needed for catalyst development and is resilient to changes in campaign goals and knowledge gaps. The discovery of new catalyst materials typically begins with limited knowledge and broad goals, lacking identified descriptors to investigate or certainty of significant design factors. After identifying initial factors, such as catalyst composition, temperature, and feed composition, a design space is defined. Initially, we start with a Base Design which is chosen as a screening design with minimal budget, which robustly identifies main effects of the factors and with which informed decisions about possible future Adaptive Designs can be based upon. By nature of the multivariate DoE approach factor effects can be estimated more efficiently and have stronger statistical significance in contrast to OFAT methods.[2] Leveraging insights gained from the Base Design, descriptors that optimally align with current objectives can be designed, while model-based approaches can subsequently quantify and predict the impact of factors on these descriptors. Based on this, unwanted factors can be eliminated after the Base Design and therefore do not waste resources in further Adaptive Designs.

Systematic Exploration of a Multi-Promoter Catalyst Composition Space with Limited Experiments: Non-Oxidative Propane Dehydrogenation to Propylene

In collaboration with BasCat (catalyst synthesis and performance evaluation) we applied this method to investigate a six-promoter design space for the supported Pt catalyzed non-oxidative propane dehydrogenation. We set out to investigate catalyst

activity (Y_2) and stability (D_2) in order to find promising new catalyst materials. High-throughput, reproducible synthesis coupled with parallel testing allowed for the evaluation of eight catalysts per week, making it costly. The descriptor map of our Base Design (Figure 1a) and model predictions of the whole design space shows that our current design space is cut off from competitive descriptor regions by a Pareto front. After conducting a detailed investigation of our model, we identified two promising promoters, which improved D_2 and Y_2. After defining a less restricted design space for these promoters, we measured an Adaptive Design (Figure 1b) for it. As a result, we were able to break through the Pareto front into more competitive descriptor regions.

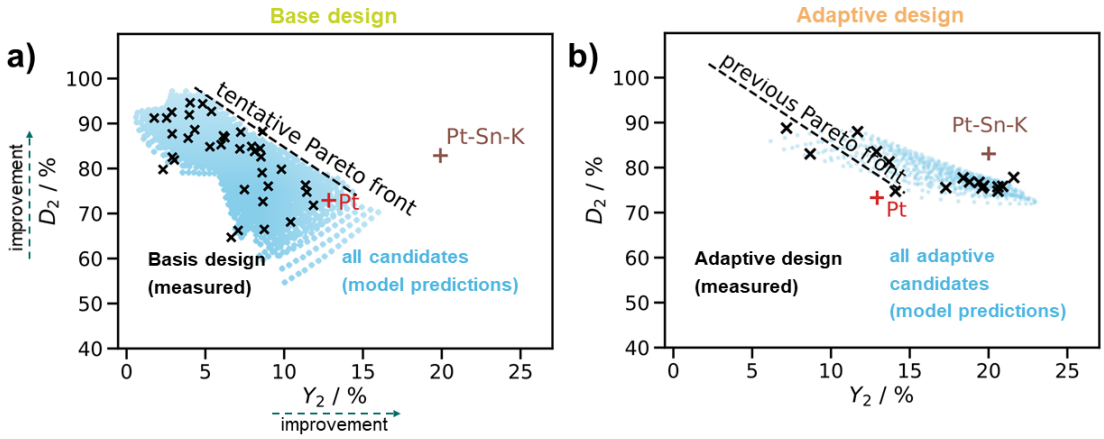

Figure 1. Overview over the multipromoter catalyst design space. (a) Performance descriptors for reaction yield Y_2 and deactivation D_2 for the 7265 unique catalysts in the design space as predicted by a regression model (blue dots). The descriptors for the 35 explicitly measured compositions of the base design are shown as black crosses. (b) Performance descriptors for the 607 unique catalysts in the design space as predicted by a regression model (blue dots). The descriptors for the 16 explicitly measured compositions of the Adaptive Design are shown as black crosses. Also shown in (a) and (b) are the descriptors for the unpromoted Pt catalyst (red cross) and the Pt–Sn–K reference catalyst (brown cross).[2]

By employing this method, we identified a novel catalyst composition within 8 weeks of experimentation that without undergoing any optimization processes demonstrates performance comparable to the highly optimized industrial catalyst Pt-Sn-K.

References:

[1] G. A. Ozin, J. Y. Y. Loh, Energy Materials Discovery: Enabling a Sustainable Future, Royal Society of Chemistry, Cambridge, 2022.

[2] C. Kunkel, F. Rüther, F. Felsen, C. W. P. Pare, A. Terzi, R. Baumgarten, E. Gioria, R. Naumann d'Alnoncourt, C. Scheurer, F. Rosowski, and K. Reuter, *ACS Catal.* 2024, **14**, 11, 9008–9017.